GGG-P-833a
January 14, 1966

SUPERSEDING
Int. Fed. Spec. GGG-P-00833 (GSA-FSS)
August 11, 1964 and
Fed. Spec. GGG-P-831c (In Part)
April 1, 1963
(See 6.2)

FEDERAL SPECIFICATION

PUNCHES, CUTTING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand punches commonly used by saddlers and metalworkers.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type, classes, and styles.* The punches shall be of the following type, classes, and styles as specified (see 6.1):

Type I—Cutting (drive).
  Class A—Rectangular (bag).
  Class B—Circular.
    Style 1—Double bow.
    Style 2—Single bow.
  Class C—Tinners'.
    Style 1—Hollow.
    Style 2—Solid.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

KK-L-258—Leather, Cattlehide, Skirting, Vegetable-Tanned.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of punches which are otherwise in accordance with this specification.

3.2 Material. Punches shall be of high-grade tool or alloy steel, properly hardened to produce punches having maximum resistance to breakage or deformation and complying with the requirements specified herein for each type, class, and style. All material shall be of uniform quality, free from all defects or imperfections that may adversely affect their serviceability and durability. Where tables list body cross section as minimum, stock sizes that are undersize within commercial tolerances for bar stock are acceptable.

3.3 Finish. Punches shall be free from forge flash, oxide scale, rust, fins, sharp edges or corners (except cutting edges), pits, burrs, cracks, and other detrimental defects. Surfaces not usually ground or polished shall have a surface finish conforming to good manufacturing practices.

3.4 Marking. Punches shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.5 Sets. When sets of punches are specified (see 6.1), all punches contained in the set shall conform to the requirements of their respective class and style covered by this specification.

3.5.1 *Containers for sets.* Punches, when ordered in encased sets, shall be furnished in durable wooden boxes, fabric wraps or rolls, or plastic containers as specified (see 6.1). The containers shall be provided with a compartment for each individual punch in order to facilitate selection, prevent damage to the punches, and provide a method of storage while the punches are not in use. All containers shall be marked plainly and legibly with the respective set designation.

3.5.1.1 *Wooden boxes.* Wood boxes shall be round or square in shape and provided with a suitable cover. The box and its cover shall be of suitable hardwood such as maple, ash, hickory, or birch. The wooden box shall have an oil finish on the inside and a varnished or shellacked finish on the outside.

3.5.1.2 *Fabric wraps or rolls.* Fabric wraps or rolls shall be of good quality, of durable material, and shall be provided with durable ties.

3.5.1.3 *Plastic containers.* Plastic containers shall be of a good quality plastic or plastic-coated cloth. The containers shall be provided with snap fasteners or durable ties. Plastic containers shall be capable of withstanding the test specified in 4.4.4.

3.6 Type I, cutting (drive). Type I punches shall be designed to have a suitable punch end and a properly proportioned shank. Classes A and B punches shall be suitable for punching holes in leather and other similar materials. Class C punch shall be suitable for punching holes in soft sheet metal. Classes A and B punches shall be capable of withstanding the test specified in 4.4.1. Class C punches shall be capable of withstanding the test specified in 4.4.2. Classes A, B, and C punches shall have, within a minimum of 1/2 inch from the cutting end, a Rockwell C hardness of from 50 to 59, or a hardness of 85.5 to 89.5 on the Rockwell superficial 15 N scale when tested as specified in 4.4.3.

3.6.1 *Punch end.* Class B punch shall be designed to provide a smooth passage for the punchings through the barrel without difficulty, and be discharged automatically. The barrel, adjacent to the cutting edge, shall be smoothly finished and polished.

3.6.1.1 *Cutting edge.* The cutting edge of the punches shall not become dulled when

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TABLE I. *Type I, class A, cutting (drive) punch, rectangular (bag)*

Size number	Dimension of hole		Length, overall ± 15 percent	Size number	Dimension of hole		Length, overall ± 15 percent
	Length	Width			Length	Width	
	Inch	*Inch*	*Inches*		*Inches*	*Inch*	*Inches*
1	1/4	5/64	4	8	1-1/8	11/64	5
2	3/8	3/32	4-3/4	9	1-1/4	11/64	5
3	1/2	7/64	4-13/16	10	1-3/8	3/16	5
4	5/8	1/8	4-7/8	11	1-1/2	13/64	5
5	3/4	9/64	5	12	1-3/4	7/32	5
6	7/8	5/32	5	13	2	1/4	5
7	1	5/32	5				

subjected to the test specified in 4.4.1 for classes A and B, and 4.4.2 for class C.

3.6.2 *Shank.* The shank shall be either circular, square, rectangular, hexagonal, or octagonal in shape (taper optional). The striking-end face shall be flat or domed, smoothly finished and at right angles to the longitudinal axis of the punch within 2°.

3.6.3 *Class A, rectangular (bag).* Class A punches shall have a rectangular-shaped cutting edge having semicircular ends, shall conform to table I, and be similar to figure 1.

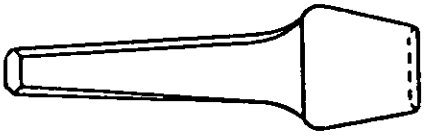


FIGURE 1. Type I, class A, cutting (drive) punch, rectangular (bag).

3.6.4 *Class B, circular.* Class B punches shall have a circular-shaped cutting edge and shall be forged of high-grade tool or alloy steel. The design of class B punches shall be such that the user may easily see the template or layout lines when the punch is positioned over the work for use.

3.6.4.1 *Style 1, double bow.* Style 1 punches shall be of the double bow, arch type and shall be either forged in one piece or a steel shank shall be welded to the punch end at the point where cuttings are discharged. The shank shall be concentric to the cutting edge with 1/16 inch for punches one inch and smaller and 1/8 inch for punches larger than one inch. Style 1 punches shall conform to table II and be similar to figure 2.

TABLE II. *Type I, class B, style 1, cutting (drive) punch, circular, double bow*

Size number	Diameter of hole	Length, overall ± 15 percent	Size number	Diameter of hole	Length, overall ± 15 percent
	Inches	*Inches*		*Inches*	*Inches*
3	3/8	5	17	1-3/8	6-1/2
4	7/16	5	18	1-1/2	6-1/2
5	1/2	5	19	1-5/8	6-1/2
6	9/16	5	20	1-3/4	7
7	5/8	5	21	1-1/8	7
8	11/16	5	22	2	7
9	3/4	5	23	2-1/8	7
10	13/16	5-1/2	24	2-1/4	7
11	7/8	5-1/2	25	2-3/8	7-1/4
12	15/16	5-1/2	26	2-1/2	7-1/4
13	1	5-1/2	27	2-3/4	7-3/4
14	1 1/16	6	28	3	7-3/4
15	1 1/8	6	29	3-1/4	8
16	1 1/4	6	30	4	8

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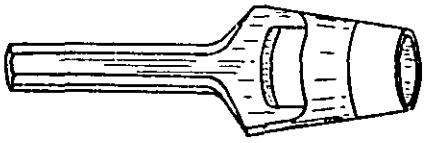


FIGURE 2. Type I, class B, style 1, cutting (drive) punch, circular, double bow.

3.6.4.2 *Style 2, single bow.* Style 2 punches shall be of the single bow type, commonly known as belt or saddlers' punch, shall conform to table III and be similar to figure 3.

TABLE III. *Type I, class B, style 2, cutting (drive) punch, circular, single bow*

Size number	Diameter of hole	Length, overall ± 15 percent
	Inch	*Inches*
1	1/16	3-1/2
2	5/64	3-1/2
4	7/64	3-1/2
5	1/8	3-3/4
6	9/64	3-3/4
7	5/32	3-3/4
8	11/64	3-3/4
9	3/16	4
11	7/32	4
13	1/4	4
15	9/32	4
16	5/16	4
18	23/64	4
19	13/32	4

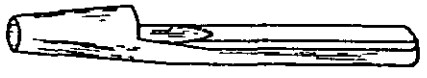


FIGURE 3. Type I, class B, style 2, cutting (drive) punch, circular, single bow.

3.6.5 *Class C, tinners'.* Class C punch shall be suitable for punching holes in soft sheet metal.

3.6.5.1 *Style 1, hollow.* Style 1 punch shall have the cutting portion circular in shape. The shank shall be of sufficient size to form a rigid tool and shall be either circular, square, rectangular, hexagonal, or octagonal in cross section. The circular cutting portion for punches of hole diameters of 5/8 inch and larger shall form a shoulder at approximately one-fourth of the length of the punch. The depth of the inside diameter of the cutting end shall be compatible with the size of the punch. The circular cutting portion of punches of hole diameters of 1/2 inch and less shall be approximately one-fourth of the length of the punch. The cutting portion of the punches shall be concentric with the axis of the punch showing a runout of not more than 0.010-inch total indicator reading. Style 1 punches shall conform to table IV and be similar to figure 4.

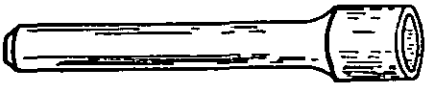


FIGURE 4. Type I, class C, style 1, cutting (drive) punch, tinners', hollow.

3.6.5.2 *Style 2, solid.* Style 2 punches shall have a circular point and a body octagonal, circular, rectangular, hexagonal, or square in cross section. The taper from the point to the body shall be of such design and length as to insure a rigid tool. The striking end and the taper shall be concentric with the axis of the body showing a runout of not more than 0.020-inch total indicator reading. The striking end and the taper shall

TABLE IV. *Type I, class C, style 1, cutting (drive) punch, tinners', hollow*

Size number	Diameter of hole	Length, overall ± 15 percent	Size number	Diameter of hole	Length, overall ± 15 percent
	Inch	*Inches*		*Inches*	*Inches*
1	1/4	4	9	1-1/4	5-1/2
2	5/16	4	10	1-1/2	6
3	3/8	4	11	1-3/4	6
4	1/2	4	12	2	6
5	5/8	5	13	2-1/4	6-1/2
6	3/4	5	14	2-1/2	6-1/2
7	7/8	5	15	2-3/4	6-1/2
8	1	5	16	3	6-1/2

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have a smoothly finished surface. Style 2 punches shall conform to table V and be similar to figure 5.

TABLE V. *Type I, class C, style 2, cutting (drive) punch, tinners', solid*

Size number	Diameter of point	Body size, across flats, cross section ± 15 percent	Length, overall ± 15 percent
	Inch	*Inch*	*Inches*
1	1/8	5/16	4-1/2
2	5/32	5/16	4-1/2
3	3/16	5/16	4-1/2
4	7/32	3/8	4-1/2
5	1/4	3/8	4-1/2
6	9/32	3/8	4-3/4
7	5/16	7/16	4-3/4
8	11/32	7/16	5
9	3/8	1/2	5

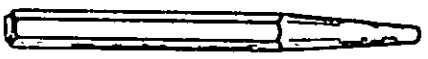


FIGURE 5. Type I, class C, style 2, cutting (drive) punch, tinners', solid.

3.7 Workmanship. Workmanship shall be first class throughout.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table VI.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.3.3 *Examination of container and sets.* Each sample unit shall be examined for any nonconformance with 3.5.1.

TABLE VI. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	2.5	Defects per 100 units	4.3.1
Dimensional examination	do.	S-3	2.5	do.	4.3.2
Examination of containers and sets*	do.	S-2	6.5	do.	4.3.3
Testing	do.	S-2	1.5	Percent defective	4.4.1 thru 4.4.4
Preparation for delivery	do.	S-2	4.0	Defects per 100 units	4.5

*Note: Punch set. — Each item in the set, including container, shall be considered as an individual item and tested accordingly.

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4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3, and 4.4.4.

4.4.1 *Performance test for cutting (drive) punch, classes A and B, styles* 1 *and* 2. Each sample punch shall be tested by punching at least 10 holes in 1/4-inch thick extra heavy leather in accordance with KK-L-258. The leather shall be placed on a suitable hardwood backing. The cutting edge shall show ease in cutting and shall demonstrate the ability of the steel to hold a keen edge under the above conditions. Attention shall be given to each sample punch as to whether the punchings pass through and are readily discharged from the barrel, where applicable, and as to whether the barrel, shank, and the striking end of the sample punch offer resistance to breakage, deformation, spalling, or cracking.

4.4.2 *Performance test for cutting (drive) punch, class C, styles* 1 *and* 2. The sample punches shall be tested by punching at least 100 holes in soft low carbon sheet steel of approximately 0.026 inch thickness. The cutting edge shall show ease in cutting, and demonstrate the ability to hold a keen edge. Attention shall be given to each sample punch as to whether the striking end, shank, body, and taper offer proper resistance to breakage, deformation, spalling, or cracking.

4.4.3 *Hardness test.* The sample punches shall be tested in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with the hardness requirement of 3.6.

4.4.4 *Flammability test for plastic wrap or roll.* The plastic wrap or roll shall be rolled into a tight roll and clamped in a support, outward in a horizontal position. There shall be clamped in a horizontal position a 5-inch square piece of 10-mesh Bunsen burner gauze, one-quarter inch below the bottom of the roll and one-half inch back from the free end of the roll. A Bunsen burner with an efficient flame 1/2 to 3/4 inch in height shall be adjusted so that the flame tip will contact the specimen when placed under the free end of the roll. At the end of 30 seconds, the flame shall be removed and the sample allowed to burn. The time required for the flame to travel from the end of the roll to the clamping device after the removal of the Bunsen burner shall determine the rate of travel of the flame. In case the roll does not continue to burn after the first ignition, the Bunsen burner shall be replaced under the free end of the roll for a period of 30 seconds immediately following the extinction of the flame on the roll. If the roll stops burning before the flame has reached the clamping device after the second ignition, the sample shall be considered self-extinguishing. When tested as specified, the plastic roll shall not burn at a rate exceeding 4 inches per minute.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table VII. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

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TABLE VII. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the punches shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.1).

5.1.1 As part of the requirements of section 5, table VIII is considered as part of table I of MIL-H-15424.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, each interior and exterior package shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, or herein, unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class and style of punches required (see 1.2).

(*c*) Size number of punches required (see applicable tables).

(*d*) Whether punches in sets are required (see 3.5).

(*e*) Whether sets should be furnished in wooden boxes, fabric wraps or rolls, or plastic containers (see 3.5.1.1, 3.5.1.2, or 3.5.1.3).

(*f*) Level of preservation, packaging, and packing required (see 5.1).

6.2 Supersession data. The classification of punches covered in this specification cor-

TABLE VIII.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Punches; cutting (drive)						
Rectangular (bag)	P-2, P-6	1	12	6	4	288
Circular	P-2, P-6	1	12	6	4	288
Double bow	P-2, P-6	1	1	12	12	144
Single bow	P-2, P-6	1	1	12	12	144
Tinners'						
Hollow	P-2, P-6	1	12	6	4	288
Solid	P-1	1	12	6	4	288

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responding to those covered in Federal Specification GGG-P-831c are as follows:

GGG-P-831c	GGG-P-00833 (GSA-FSS)
Type VI—Cutting (drive).	Type I—Cutting (drive).
Class A—Rectangular (bag).	Class A—Rectangular (bag).
Class B—Circular.	Class B—Circular.
Style 1—Double bow.	Style 1—Double bow.
Style 2—Single bow.	Style 2—Single bow.
Class C—Tinners'.	Class C—Tinners'.
Style 1—Hollow.	Style 1—Hollow.
Style 2—Solid.	Style 2—Solid.

6.3 Transportation descriptions. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, other than power.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

DOD coordination has been waived.

Preparing Activity: GSA-FSS

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